

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	HDI Predictor Using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution uses Machine Learning to predict the Human Development Index (HDI), delivering fast, accurate, and data-driven insights from key socio-economic indicators. By integrating a trained Linear Regression model into a Flask-based web application, the system supports quicker and more informed decision-making.

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based web application that accurately predicts a country's Human Development Index (HDI) using key socio-economic indicators, enabling faster analysis and better-informed decisions.
Scope	The project covers collecting and preprocessing HDI-related data, training a Linear Regression model, and deploying the trained model through a Flask web application to deliver real-time HDI predictions.
Problem Statement	
Description	Manual HDI analysis involves evaluating multiple socio-economic indicators, making the process time-consuming and potentially delaying decision-making.

Impact	An automated HDI prediction system enables researchers, students, and policy makers to obtain quick and reliable predictions, supporting improved planning and development analysis.
Proposed Solution	
Approach	Develop a Linear Regression-based Machine Learning model to identify relationships between socio-economic indicators and HDI. The trained model will be deployed using Flask to provide an interactive web interface for prediction.
Key Features	Machine Learning-based HDI prediction using Linear Regression.



	- Interactive Flask web application for user input. - Predicts the HDI score and corresponding HDI category using a pre-trained HDI.pkl model, with a simple user interface and real-time prediction without retraining.
--	---

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing Resources	Intel Core i5 or above
Memory	Memory	8 GB RAM
Storage	Storage	256 GB SSD or higher
Software		

Framework	Framework	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Development Environment	IDE	VS Code, Jupyter Notebook
Data		
Dataset	Dataset	HDI.csv (Human Development Index Dataset)